

LESSON 3-5 • SESSION 2 • PRACTICE • FORM A

Comparing Rates with Benchmarks

Learning target – I can use a benchmark to compare two rates without dividing. • 6.RP.A.3b

REMEMBER

- A **benchmark** is one amount you use for **both** choices. Scale each rate to it, then compare.
- Pick a benchmark both amounts divide into evenly: 12 works for 4 and 6.
- You can also **divide** to find each **unit rate** and compare. Round money to the nearest cent.

WORKED EXAMPLE

Hose A fills 6 gallons in 2 minutes. Hose B fills 10 gallons in 4 minutes. Which is faster?

Step 1 Benchmark: 20 minutes.

Step 2 A: $6 \times 10 = 60$ gallons. B: $10 \times 5 = 50$ gallons.

Step 3 Check: $6 \div 2 = 3$ and $10 \div 4 = 2.5$ gallons per minute.

Answer Hose A is faster

WORDS benchmark (*punto de referencia*) • rate (*tasa*) • unit rate (*tasa unitaria*)

PRACTICE

show your work • label units

1. Find each unit rate.

a. 48 flyers in 3 min ____ per min

b. 84 stairs in 6 min ____ per min

c. \$15 for 6 lb ____ per lb

d. 450 miles on 15 gal ____ per gal

2. **Arcade Games.** Game A costs 2 tokens and gives 4 tickets. Game B costs 3 tokens and gives 7 tickets. Use a 30-token benchmark.

Hint Game A $\times 15$; Game B $\times 10$.

	Tokens	Tickets	Tickets for 30 tokens
Game A	2	4	
Game B	3	7	

Better game _____

3. **Typing Contest.** Tyler types 140 words in 4 minutes. Sarah types 175 words in 5 minutes. Who types faster?

Hint Find words per minute for each.

Show your work.

Faster _____

4. **Gummy Bears.** Store A sells 6 lb for \$15. Store B sells 8 lb for \$18. Use a 24-lb benchmark to find the better buy.

Show your work.

Better buy _____

5. **Corner Store Chips.** Store J sells 4 bags of chips for \$9.00. Store K sells 6 bags for \$13.20.

a. Price per bag: Store J ____ Store K ____

b. Cost of 12 bags: Store J ____ Store K ____

c. Which store is the better buy? _____

- 6. Ice Rink Passes.** Rink A: 5 sessions for \$42.50. Rink B: 8 sessions for \$66.40. Find the cost per session. Which is the better deal?

Hint Divide each price by the number of sessions.

Show your work.

Better deal _____

- 7. Reading Rates.** Reader 1 finishes 40 pages in 5 days. Reader 2 finishes 54 pages in 6 days.

Hint $30 \div 5 = 6$ and $30 \div 6 = 5$.

- a.** Pages in 30 days: Reader 1 _____ Reader 2 _____
- b.** Who reads faster? _____

Show your work.

- 8. Field Trip Vans.** Van company A charges \$120 for 8 students. Company B charges \$176 for 11 students.

- a.** Use 88 students as a benchmark. Find each total. _____
- b.** Which company is cheaper per student? _____
- c.** Why might a benchmark be easier than dividing? _____

FIND THE ERROR

explain the mistake, then fix it

- 9. Carlos's Work.** Ride A costs 2 tickets for 9 points. Ride B costs 5 tickets for 15 points. Carlos subtracted $9 - 2 = 7$ and $15 - 5 = 10$, so he picked Ride B. Explain his mistake and find the better ride.

TEST PRACTICE

choose the best answer

- 10.** A good benchmark for comparing a 4-pack and a 6-pack is –
- A** 10 **B** 12 **C** 16 **D** 18
- 11.** At a 12-unit benchmark, Store X costs \$30 and Store Y costs \$27. The better buy is –
- A** Store X **B** Store Y
- C** both the same **D** impossible to tell

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

270 R	\$8.30 A	\$54 M	\$0.12 Q	70 O	\$26.40 P	16 C	\$1,408 E	\$34 Z	324 X
#1a		#2	#4	#5b	#6	#7a	#8a		

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #_____